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List of Participants of the Discussion Meeting

One-Day Discussion Meeting on

A Primer on Rapid Visual Screening (RVS)

Consolidating Earthquake Safety Assessment Efforts in India

9 January 2020 at IIIT Hyderabad

List of Participants

Government

1. Sri Kamal Kishore, Member, NDMA, Government of India, New Delhi
2. Sri Sandeep Poundrik, JS (Mitigation), NDMA, Government of India, New Delhi
3. Sri Mahendra Meena, Sr. Consultant (Earthquake & Tsunami), NDMA, Government of India, New Delhi
4. Sri Javed Iqbal, Jr. Consultant (Earthquake), NDMA, Government of India, New Delhi

Academia

5. Hemant B. Kaushik, Associate Professor, IIT Guwahati
6. Goutam Mondal, Assistant Professor, IIT Bhubaneswar
7. Rupen Goswami, Associate Professor, IIT Madras
8. C. V. R. Murty, Professor, IIT Madras
9. Dipti Ranjan Sahoo, Associate Professor, IIT Delhi
10. Ravi Sinha, Professor, IIT Bombay (through Skype)
11. Sunitha Palissery, Assistant Professor, IIIT Hyderabad
12. R. Pradeep Kumar, Professor, IIIT Hyderabad
13. Hemant Kumar Vinayak, Assistant Professor, NIT, Hamirpur
14. O. R. Jaiswal, Professor, VNIT, Nagpur
15. Surendranadh Somala, Assistant Professor, IIT Hyderabad

R&D

16. Srinagesh Davuluri, Head, Seismology Laboratory, CSIR NGRI, Hyderabad
17. R. K. Chadha, Raja Ramanna Fellow, CSIR NGRI, Hyderabad
18. Ajai P. Chaurasia, Chief Principal Scientist, CBRI, Roorkee

Industry

19. Hari Kumar, Managing Director, Geohazards Society, New Delhi
20. Arvind K. Jaiswal, Chief Consultant, EON Designers, Secunderabad
21. Jaswant N. Arlekar, Principal Consultant, M/s Mandrekar & Associates, Mumbai

Annexure B

Recommended Pre and Post Rapid Earthquake Assessment Forms for:

1. Reinforced Concrete Building
2. Burnt Clay Bricks Building
3. Confined Masonry Building
4. Random Rubble Masonry Building
5. Mud House
6. Dhajji Dewari
7. Ekra House

A: RC

Pre-Earthquake Rapid Visual Screening
Reinforced Concrete Buildings

Inspection

Inspector ID:
Inspection Date :
Inspection Time:

Building Description

Building Name :
Address 1 :
Address 2 :
City/Town :
Coordinates : N_____° E_____°

Structural System

Structural System

☐ *Moment Frame* ☐ *Moment Frame with Braces*
☐ *Moment Frame with Structural Walls* ☐ *Other_____*

Structural Components

FLOOOR ☐ *In-situ* ☐ *Precast Planks with in-situ screed* ☐ *Precast* ☐ *Other_____*
ROOF ☐ *Flat* ☐ *Pitched* ☐ *Hipped* ☐ *Split* ☐ *Other_____*

Occupancy

RESIDENTIAL	*Individual House*	☐ *Apartments*	
EDUCATIONAL	☐ *School*	☐ *College*	☐ *Institute/University*
LIFELINE	☐ *Hospital*	☐ *Police Station*	☐ *Fire Station*
	☐ *Power Station*	☐ *Water Plant*	☐ *Sewage Plant*
COMMERCIAL	☐ *Hotel*	☐ *Shopping*	☐ *Recreational*
OFFICE	☐ *Government*	☐ *Private*	
MIXED USE	☐ *Residential and Commercial*		☐ *Residential and Industrial*
INDUSTRIAL	☐ *Agriculture*	☐ *Live Stock*	
OTHER	☐ _____		

S.No.	Life Threatening Parameters		Tag
1.	**Siting Issues**	1.1 Building is built on hill slope or adjacent to hill slope, that is vulnerable to falling debris.	**Red**
		1.2 Building rests adjoining a severely deteriorated or damaged building or structure.	**Red**
2.	**Soil & Foundation Conditions**	2.1 Building is built on liquefiable soil.	**Red**
		2.2 Building is built on river terraces that can slide or creep.	**Red**
		2.3 Building is built on hill slope that can slide.	**Red**
3.	**Architectural Features**	3.1 Building has exterior plan dimension smaller at the plinth level than at upper storeys.	**Red**
		3.2 Building has heavier upper storeys.	**Red**
		3.3 Building has large unanchored projections or overhangs.	**Red**
		3.4 Building has an Open Storey (at ground /other level) with slender columns.	**Red**
		3.5 Building has RC frame with Floating Columns.	**Red**
		3.6 Building has irregular plan geometry and or plan aspect ratio > 4.0.	**Red**

4.	**Structural Aspects**	4.1 Building has Slender Columns and Stiff Beams, with glass infills or no infills.	**Red**
		4.2 Building has Flat Slabs without any RC Structural Walls in each plan direction.	**Red**
5.	**Siting Issues**	5.1 Building is resting on cracked ground.	**Yellow**
		5.2 Building is adjoining another building on the side with no gap.	**Yellow**
6.	**Soil & Foundation Conditions**	6.1 Building is resting on weak or non-uniform soil along the length in plan.	**Yellow**
7.	**Architectural Features**	7.1 Building has large or heavy cantilever parapet walls.	**Yellow**
		7.2 Building has large or heavy cantilever chimneys.	**Yellow**
		7.3 Building has large or heavy cantilever balconies or sunshades.	**Yellow**
		7.4 Building has large room sizes and large storey heights.	**Yellow**
		7.5 Building has windows larger than 50% of the length of the wall.	**Yellow**
8.	**Structural Aspects**	8.1 Building has an irregular structural grid.	**Yellow**
		8.2 Building has RC walls unsymmetrical in both directions.	**Yellow**
		8.3 Building has staircase at unsymmetrical location in plan.	**Yellow**
		8.4 Building has unanchored water tanks on roof.	**Yellow**
9.	**Construction and Maintenance Details**	9.1 Building has longitudinal cracks and/or spalling in Beams and Columns.	**Yellow**
		9.2 Building has poor construction quality material.	**Yellow**
10.		None of the above.	Green

FINAL RATING

GREEN (Usable)	**YELLOW (Usable with temporary interventions)**	**RED (Unusable)**
	Suggested interventions:	

Further Actions

Building to be quarantined : ☐ Not Recommended ☐ Recommended

Level 2 Detailed Visual Screening : ☐ Not Recommended ☐ Recommended

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B: RC

Post-Earthquake Rapid Visual Screening
Reinforced Concrete Buildings

Inspection

Inspector ID:
Inspection Date :
Inspection Time:

Building Description

Building Name :
Address 1 :
Address 2 :
City/Town :
Coordinates : N_____° E_____°

Structural System

Structural System

☐ *Moment Frame* ☐ *Moment Frame with Braces*
☐ *Moment Frame with Structural Walls* ☐ *Other_____*

Structural Components

FLOOOR ☐ *In-situ* ☐ *Precast Planks with in-situ screed* ☐ *Precast* ☐ *Other_____*
ROOF ☐ *Flat* ☐ *Pitched* ☐ *Hipped* ☐ *Split* ☐ *Other_____*

Occupancy

RESIDENTIAL	☐ *Individual House*	☐ *Apartments*	
EDUCATIONAL	☐ *School*	☐ *College*	☐ *Institute/University*
LIFELINE	☐ *Hospital*	☐ *Police Station*	☐ *Fire Station*
	☐ *Power Station*	☐ *Water Plant*	☐ *Sewage Plant*
COMMERCIAL	☐ *Hotel*	☐ *Shopping*	☐ *Recreational*
OFFICE	☐ *Government*	☐ *Private*	
MIXED USE	☐ *Residential and Commercial*		☐ *Residential and Industrial*
INDUSTRIAL	☐ *Agriculture*	☐ *Live Stock*	
OTHER	☐ _____		

S.No.	*Life Threatening Parameters*		*Tag*
1.	***Siting Issues***	*1.1 Building is resting on ground that has failed due to Landslide/Fissures and Liquefaction.*	***Red***
		1.2 Building is tilted.	***Red***
		1.3 Building is resting on hill slopes or adjacent to hill slopes, and has unsafe/tilted adjoining/uphill building or loose boulders.	***Red***
2.	***Soil & Foundation Conditions***	*2.1 Building is resting on river terraces that have cracked soil.*	***Red***
		2.2 Building has sustained uneven settlement of the ground.	***Red***
		2.3 Building is resting on liquefied soil.	***Red***
		2.4 Uneven Settlement of building leading to visual damage.	***Red***
3.	***Structural Aspects***	*3.1 Building has pounding damage from adjoining building or structure.*	***Red***
		3.2 Building has collapsed/damaged staircase/mumty or blockade of staircase.	***Red***
		3.3 Building has an Open storey (at ground /other level) with columns having shear cracks.	***Red***
		3.4 Building has Floating Columns with cracked supporting beams.	***Red***
		3.5 Building has main load resisting columns with shear cracks.	***Red***
		3.6 Building has main load resisting short columns with shear cracks.	***Red***
		3.7 Building has Flat Slab with punching shear failure.	***Red***

		3.8 *Building has spalling of cover concrete in main load resisting columns.*	***Red***
		3.9 *Building has extensive X cracking or out-of-plane collapse of infills.*	***Red***
		3.10 *No seismic separation between staircase and building*	***Red***
4.	***Siting Issues***	4.1 *Building is adjacent to a failed slope.*	***Yellow***
5.	***Structural Aspects***	5.1 *Building has sustained collapse of cantilevers, balconies, chimneys and parapets.*	***Yellow***
		5.2 *Building has water tanks on roof displaced from their supports.*	***Yellow***
		5.3 *Building has sustained extensive separation between infill wall and frame members, and has no damage in columns.*	***Yellow***
6.		*None of the above.*	*Green*

FINAL RATING		
GREEN (Usable)	***YELLOW (Usable with temporary interventions)***	***RED (Unusable)***
	Suggested interventions:	

Further Actions

Building to be quarantined : ☐ *Not Recommended* ☐ *Recommended*

Level 2 Detailed Visual Screening : ☐ *Not Recommended* ☐ *Recommended*

A : BCBM

Pre-Earthquake Rapid Visual Screening
Burnt Clay Brick Masonry Buildings

Inspection

Inspector ID:
Inspection Date :
Inspection Time:

Building Description

Building Name :
Address 1 :
Address 2 :
City/Town :
Coordinates : N_____° E_____°

Structural Components

STRUCTURAL SYSTEM	☐	*Load Bearing Masonry Structure*					☐	*Other*_____
FLOOR	☐	*RC Slab*	☐	*Precast Planks with In-situ screed*			☐	*Other*_____
ROOF GEOMETRY	☐	*Pitched*	☐	*Hipped*	☐	*Split*	☐	*Other*_____
ROOF MATERIAL	☐	*RC Slab*	☐	*Wood with Clay tiles*	☐	*Wood truss with Corrugated sheets*		
			☐	*Wood with Wooden Planks*			☐	*Other*_____
MORTAR	☐	*Cement*	☐	*Mud*	☐	*Lime*	☐	*No mortar*	☐	*Other*_____

Occupancy

RESIDENTIAL	☐ *Individual House*	☐ *Apartments*		
EDUCATIONAL	☐ *School*	☐ *College*	☐ *Institute/University*	
LIFELINE	☐ *Hospital*	☐ *Police Station*	☐ *Fire Station*	
	☐ *Power Station*	☐ *Water Plant*	☐ *Sewage Plant*	
COMMERCIAL	☐ *Hotel*	☐ *Shopping*	☐ *Recreational*	
OFFICE	☐ *Government*	☐ *Private*		
MIXED USE	☐ *Residential and Commercial*		☐ *Residential and Industrial*	
INDUSTRIAL	☐ *Agriculture*	☐ *Live Stock*		
OTHER	☐ _____			

S.No.	Life Threatening Parameters			Tag
1.	**Siting Issues**	1.1	*Building is built on hill slope or adjacent to hill slope, that is vulnerable to falling debris.*	**Red**
		1.2	*Building rests adjoining a severely deteriorated or damaged building or structure.*	**Red**
2.	**Soil and foundation conditions**	2.1	*Building is built on liquefiable soil.*	**Red**
		2.2	*Building is built on river terraces that can slide or creep.*	**Red**
		2.3	*Building is built on hill slope that can slide.*	**Red**
3.	**Architecture Features**	3.1	*Building has exterior plan dimension smaller at the plinth level than at upper storeys.*	**Red**
		3.2	*Building has heavier upper storeys.*	**Red**
		3.3	*Building has large unanchored projections or overhangs.*	**Red**
		3.4	*Building has irregular plan geometry and/or plan aspect (length-to-width ratio) >4.0.*	**Red**
		3.5	*Building has storey height more than about 25 times wall thickness.*	**Red**
		3.6	*Building has unsupported height more than 2.5m.*	**Red**
		3.7	*Building has stiff upper storeys.*	**Red**
4.		4.1	*Building has no continuous lintel band, roof band and gable band.*	**Red**

	Structural Aspects	4.2 Building has door/window openings in walls close to corners.	**Red**
		4.3 Building has masonry walls not integrated into each other at corners.	**Red**
		4.4 Building has walls made without lime / cement mortar or mud mortar exposed to vagaries.	**Red**
		4.5 Building has roof not anchored into the walls.	**Red**
		4.6 Building has thick walls made of two wythes and no through stones.	**Red**
		4.7 Building has out-of-plumb walls.	**Red**
		4.8 Building has separation between walls and between walls and floor/roof.	**Red**
		4.9 Building has cracks in walls.	**Red**
5.	**Construction and Material details**	5.1 Buildings have walls made with mud mortar and deteriorated significantly owing to vagaries of the outside weather (especially rainwater beating).	**Red**
6.	**Siting Issues**	6.1 Building is resting on cracked ground.	**Yellow**
		6.2 Building is adjoining another building on the side with no gap.	**Yellow**
		6.3 Building is resting on sloped ground with access to house at two/three levels, i.e., ground, middle floor and roof.	**Yellow**
		6.4 Building is connected to the sloped ground and there is no gap between the building and the natural slope of the ground.	**Yellow**
7.	**Soil and foundation conditions**	7.1 Building is resting on weak or non-uniform soil along the length in plan.	**Yellow**
8.	**Architectural Features**	8.1 Building has large or heavy cantilever parapet walls.	**Yellow**
		8.2 Building has large or heavy cantilever chimneys.	**Yellow**
		8.3 Building has large or heavy cantilever balconies or sunshades.	**Yellow**
		8.4 Building has large room sizes and large storey heights.	**Yellow**
		8.5 Building has windows larger than 50% of the length of the wall.	**Yellow**
		8.6 Building has unanchored water tanks on roof.	**Yellow**
9.	**Structural Aspects**	9.1 Building has an irregular grid of masonry walls in plan.	**Yellow**
		9.2 Building has staircase at unsymmetrical location in plan.	**Yellow**
		9.3 Building has loosely placed country tiles on the roof.	**Yellow**
10	**Construction and Material details**	10.1 Building has vertical cracks in masonry walls.	**Yellow**
		10.2 Building has poor construction quality material.	**Yellow**
11		None of the above.	**Green**

FINAL RATING

GREEN (Usable)	**YELLOW (Usable with temporary interventions)**	**RED (Unusable)**
	Suggested interventions:	

Further Actions

Building to be quarantined : ☐ Not Recommended ☐ Recommended

Level 2 Detailed Visual Screening : ☐ Not Recommended ☐ Recommended

.

B : BCBM

Post-Earthquake Rapid Visual Screening
Burnt Clay Brick Masonry Buildings

Inspection

Inspector ID:
Inspection Date :
Inspection Time:

Building Description

Building Name :
Address 1 :
Address 2 :
City/Town :
Coordinates : N_____° E_____°

Structural Components

STRUCTURAL SYSTEM	☐	*Load Bearing Masonry Structure*					☐	*Other_____*
FLOOR	☐	*RC Slab*	☐	*Precast Planks with In-situ screed*			☐	*Other_____*
ROOF GEOMETRY	☐	*Pitched*	☐	*Hipped*	☐	*Split*	☐	*Other_____*
ROOF MATERIAL	☐	*RC Slab*	☐	*Wood with Clay tiles*	☐	*Wood truss with Corrugated sheets*		
			☐	*Wood with Wooden Planks*			☐	*Other_____*
MORTAR	☐	*Cement*	☐	*Mud*	☐	*Lime*	☐	*No mortar*	☐	*Other_____*

Occupancy

RESIDENTIAL	☐ *Individual House*	☐ *Apartments*		
EDUCATIONAL	☐ *School*	☐ *College*	☐ *Institute/University*	
LIFELINE	☐ *Hospital*	☐ *Police Station*	☐ *Fire Station*	
	☐ *Power Station*	☐ *Water Plant*	☐ *Sewage Plant*	
COMMERCIAL	☐ *Hotel*	☐ *Shopping*	☐ *Recreational*	
OFFICE	☐ *Government*	☐ *Private*		
MIXED USE	☐ *Residential and Commercial*		☐ *Residential and Industrial*	
INDUSTRIAL	☐ *Agriculture*	☐ *Live Stock*		
OTHER	☐ _____			

S.No.	Life Threatening Parameters		Tag
1.	**Siting Issues**	1.1 Building is resting on ground that has failed due to Landslide/Fissures and Liquefaction.	Red
		1.2 Building is tilted.	Red
		1.3 Building is resting on hill slopes or adjacent to hill slopes.	Red
2.	**Soil and foundation conditions**	2.1 Building is resting on river terraces that have cracked or soil.	Red
		2.2 Building has sustained uneven settlement of the ground.	Red
		2.3 Building is resting on liquefied soil.	Red
		2.4 Uneven Settlement of building leading to visual damage.	Red
3.	**Structural Aspects**	3.1 Building has pounding damage from adjoining building or structure.	Red
		3.2 Building has collapsed/damaged staircase/mumty or blockade of staircase.	Red
		3.3 Building has plinth masonry severely damaged.	Red
		3.4 Building has horizontally slid at any storey.	Red
		3.5 Building has suffered separation of wall junctions at corners.	Red
		3.6 Building has floor-wall junction separated with walls out-of-plumb.	Red
		3.7 Building has suffered gable collapse.	Red
		3.8 Building has walls separated into wythes.	Red
		3.9 Building has diagonal shear cracking in wall piers and/or spandrels.	Red

		3.10 Building has crushed masonry at wall base.	***Red***
		3.11 No seismic separation between staircase and building.	***Red***
4.	***Siting Issues***	*4.1 Building is adjacent to a failed slope.*	***Yellow***
5.	***Structural Aspects***	*5.1 Building has sustained collapse of cantilevers, balconies, chimneys and parapets.*	***Yellow***
		5.2 Building has water tanks on roof displaced from their supports.	***Yellow***
6.		*None of the above.*	***Green***

FINAL RATING		
GREEN (Usable)	***YELLOW (Usable with temporary interventions)***	***RED (Unusable)***
	Suggested interventions:	

Further Actions

Building to be quarantined : ☐ *Not Recommended* ☐ *Recommended*

Level 2 Detailed Visual Screening : ☐ *Not Recommended* ☐ *Recommended*

A: CM

Pre-Earthquake Rapid Visual Screening
Confined Masonry

Inspection

Inspector ID:
Inspection Date :
Inspection Time:

Building Description

Building Name :
Address 1 :
Address 2 :
City/Town :
Coordinates : N_____° E_____°

Structural System

Structural System

Load Bearing Masonry Structure with
- ☐ *Burnt clay bricks*
- ☐ *Cement blocks*
- ☐ *Stone blocks*
- ☐ *Others _____*

Structural Components

FLOOR ☐ *RC Slab* ☐ *Other _____*

ROOF GROMETRY ☐ *Pitched* ☐ *Hipped* ☐ *Split* ☐ *Other _____*

ROOF MATERIAL ☐ *RC Slab* ☐ *Wood Truss with*
- ○ *Clay tiles*
- ○ *Corrugated sheets*
- ○ *Wood Planks*

☐ *Other _____*

Occupancy

RESIDENTIAL	☐ *Individual House*	☐ *Apartments*	
EDUCATIONAL	☐ *School*	☐ *College*	☐ *Institute/University*
LIFELINE	☐ *Hospital*	☐ *Police Station*	☐ *Fire Station*
	☐ *Power Station*	☐ *Water Plant*	☐ *Sewage Plant*
COMMERCIAL	☐ *Hotel*	☐ *Shopping*	☐ *Recreational*
OFFICE	☐ *Government*	☐ *Private*	
MIXED USE	☐ *Residential and Commercial*		☐ *Residential and Industrial*
INDUSTRIAL	☐ *Agriculture*	☐ *Live Stock*	
OTHER	☐ _____		

S.No.	Life Threatening Parameters		Tag
1.	**Siting Issues**	1.3 *Building is built on hill slope or adjacent to hill slope, that is vulnerable to falling debris.*	**Red**
		1.4 *Building rests adjoining a severely deteriorated or damaged building or structure.*	**Red**
2.	**Soil & Foundation Conditions**	2.4 *Building is built on liquefiable soil.*	**Red**
		2.5 *Building is built on river terraces that can slide or creep.*	**Red**
		2.6 *Building is built on hill slope that can slide.*	**Red**
3.	**Architectural Features**	3.7 *Building has exterior plan dimension smaller at the plinth level than at upper storeys.*	**Red**
		3.8 *Building has heavier upper storeys.*	**Red**
		3.9 *Building has large unanchored projections or overhangs.*	**Red**
		3.10 *Building has no continuous tie beams or tie columns.*	**Red**

		3.11 Building has no Tie Beams at plinth, lintel or roof levels.	**Red**
		3.12 Building has no Tie Columns at wall junctions.	**Red**
		3.13 Building has plan aspect (length-to-width ratio) more than 4.0.	**Red**
		3.14 Building has irregular plan.	**Red**
		3.15 Building has storey height more than about ~25 times wall thickness.	**Red**
4.	**Structural Aspects**	4.1 Building has door/window openings in walls close to corners.	**Red**
		4.2 Building has no toothing between masonry walls and Tie Columns.	**Red**
		4.3 Building has Tie Columns spaced at more than 1.5 times Storeyheight.	**Red**
		4.4 Building has out-of-plumb walls.	**Red**
5.	**Siting Issues**	5.1 Building is resting on cracked ground.	**Yellow**
		5.2 Building is adjoining another building on the side with no gap.	**Yellow**
6.	**Soil & Foundation Conditions**	6.1 Building is resting on weak or non-uniform soil along the length in plan.	**Yellow**
7.	**Architectural Features**	7.1 Building has large or heavy cantilever parapet walls.	**Yellow**
		7.2 Building has large or heavy cantilever chimneys.	**Yellow**
		7.3 Building has large or heavy cantilever balconies or sunshades.	**Yellow**
		7.4 Building has large room sizes and large storey heights.	**Yellow**
		7.5 Building has windows larger than 50% of the length of the wall.	**Yellow**
		7.6 Building has Tie Columns spaced at more than Storey Height.	**Yellow**
		7.7 Building has no spandrel masonry between Tie Columns and Tie Beams above lintel.	**Yellow**
		7.8 Floors with (a) Timber planks with Timber or Steel Joists, (b) Timber Planks and Stone Slabs, and/or (c) Brick, brick coba or mud overlay.	**Yellow**
8.	**Structural Aspects**	8.1 Building has an irregular grid of masonry walls in plan.	**Yellow**
		8.2 Building has staircase at unsymmetrical location in plan.	**Yellow**
		8.3 Building has loosely placed country tiles on the roof.	**Yellow**
		8.4 Building has unanchored water tanks on roof.	**Yellow**
		8.5 Building has no anchorage of steel bar in Tie Columns into foundation.	**Yellow**
		8.6 Building has discontinuous Tie Beams at sill and lintel levels.	**Yellow**
9.	**Construction and Maintenance Details**	9.1 Building has vertical cracks in masonry walls, and/or longitudinal cracks and spalling in Tie Beams and Tie Columns.	**Yellow**
		9.2 Building has poor construction quality material.	**Yellow**
10.		None of the above.	Green

FINAL RATING

GREEN (Usable)	**YELLOW (Usable with temporary interventions)**	**RED (Unusable)**
	Suggested interventions:	

Further Actions

Building to be quarantined : ☐ Not Recommended ☐ Recommended

Level 2 Detailed Visual Screening : ☐ Not Recommended ☐ Recommended

B: CM

Post-Earthquake Rapid Visual Screening
CONFINED MASONRY

Inspection

Inspector ID:

Inspection Date :

Inspection Time:

Building Description

Building Name :

Address 1 :

Address 2 :

City/Town :

Coordinates : N_____° E_____°

Structural System

Structural System

Load Bearing Masonry Structure with
- ☐ *Burnt clay bricks*
- ☐ *Cement blocks*
- ☐ *Stone blocks*
- ☐ *Others _____*

Structural Components

FLOOR ☐ *RC Slab* ☐ *Other_____*

ROOF GROMETRY ☐ *Pitched* ☐ *Hipped* ☐ *Split* ☐ *Other_____*

ROOF MATERIAL ☐ *RC Slab* ☐ *Wood Truss with*
- ○ *Clay tiles*
- ○ *Corrugated sheets*
- ○ *Wood Planks*

☐ *Other_____*

Occupancy

RESIDENTIAL	☐ *Individual House*	☐ *Apartments*	
EDUCATIONAL	☐ *School*	☐ *College*	☐ *Institute/University*
LIFELINE	☐ *Hospital*	☐ *Police Station*	☐ *Fire Station*
	☐ *Power Station*	☐ *Water Plant*	☐ *Sewage Plant*
COMMERCIAL	☐ *Hotel*	☐ *Shopping*	☐ *Recreational*
OFFICE	☐ *Government*	☐ *Private*	
MIXED USE	☐ *Residential and Commercial*		☐ *Residential and Industrial*
INDUSTRIAL	☐ *Agriculture*	☐ *Live Stock*	
OTHER	☐ _____		

S.No.	Life Threatening Parameters		Tag
1.	**Siting Issues**	1.1 *Building is resting on ground that has failed due to Landslide/Fissures and Liquefaction.*	**Red**
		1.2 *Building is tilted.*	**Red**
		1.3 *Building is resting on hill slopes or adjacent to hill slopes, and has unsafe/tilted adjoining/uphill building or loose boulders.*	**Red**
2.	**Soil & Foundation Conditions**	2.1 *Building is resting on river terraces that have cracked or soil.*	**Red**
		2.2 *Building has sustained uneven settlement of the ground.*	**Red**
		2.3 *Building is resting on liquefied soil.*	**Red**
		2.4 *Uneven Settlement of building leading to visual damage.*	**Red**
3.	**Structural Aspects**	3.1 *Building has pounding damage from adjoining building or structure.*	**Red**
		3.2 *Building has collapsed/damaged staircase/mumty or blockade of staircase.*	**Red**

		3.3 *Building has plinth masonry severely damaged.*	**Red**
		3.4 *Building has horizontally slid at any storey.*	**Red**
		3.5 *Building has suffered separation of Tie columns and masonry wallsat corners.*	**Red**
		3.6 *Building has floor-wall junction separated with walls out-of-plumb.*	**Red**
		3.7 *Building has suffered gable collapse.*	**Red**
		3.8 *Building has through cracks across Tie Columns and Tie Beams.*	**Red**
		3.9 *Building has diagonal shear cracking in wall panels.*	**Red**
		3.10 *Building has crushed masonry at wall base.*	**Red**
		3.11 *No seismic separation between staircase and building.*	**Red**
4.	***Siting Issues***	4.1 *Building is adjacent to a failed slope.*	***Yellow***
5.	***Structural Aspects***	5.1 *Building has sustained collapse of cantilevers, balconies, chimneys and parapets.*	***Yellow***
		5.2 *Uneven settlement of adjacent building.*	***Yellow***
		None of the above.	*Green*

FINAL RATING		
GREEN (Usable)	***YELLOW (Usable with temporary interventions)***	***RED (Unusable)***
	Suggested interventions:	

Further Actions

Building to be quarantined ☐ *Not Recommended* ☐ *Recommended*

Level 2 Detailed Visual Screening ☐ *Not Recommended* ☐ *Recommended*

A: RRSM

Pre-Earthquake Rapid Visual Screening
Random Rubble Stone Masonry

Inspection

Inspector ID:

Inspection Date :

Inspection Time:

Building Description

Building Name :

Address 1 :

Address 2 :

City/Town :

Coordinates : N_____° E_____°

Structural System

Structural System

Load Bearing Masonry Structure with
- ☐ *Igneous rocks*
- ☐ *Sedimentary rocks*
- ☐ *Slate blocks*
- ☐ *Others _____*

Structural Components

FLOOR	☐ *RC Slab*	☐ *Timber planks on Timber beams*		☐ *Other_____*
ROOF GROMETRY	☐ *Pitched*	☐ *Hipped*	☐ *Split*	☐ *Other_____*
ROOF MATERIAL	☐ *RC Slab*	☐ *Wood Truss with*	○ *Clay tiles*	
			○ *Corrugated sheets*	
			○ *Wood Planks*	
	☐ *Steel Truss with corrugated sheeting*			☐ *Other_____*
MORTAR	☐ *Cement*	☐ *Mud*	☐ *Lime* ☐ *No mortar*	☐ *Other_____*

Occupancy

RESIDENTIAL	☐ *Individual House*	☐ *Apartments*	
EDUCATIONAL	☐ *School*	☐ *College*	☐ *Institute/University*
LIFELINE	☐ *Hospital*	☐ *Police Station*	☐ *Fire Station*
	☐ *Power Station*	☐ *Water Plant*	☐ *Sewage Plant*
COMMERCIAL	☐ *Hotel*	☐ *Shopping*	☐ *Recreational*
OFFICE	☐ *Government*	☐ *Private*	
MIXED USE	☐ *Residential and Commercial*		☐ *Residential and Industrial*
INDUSTRIAL	☐ *Agriculture*	☐ *Live Stock*	
OTHER	☐ _____		

S.No.	Life Threatening Parameters		Tag
1.	**Siting Issues**	1.5 *Building is built on hill slope or adjacent to hill slope, that is vulnerable to falling debris.*	**Red**
		1.6 *Building rests adjoining a severely deteriorated or damaged building or structure.*	**Red**
2.	**Soil & Foundation Conditions**	2.7 *Building is built on liquefiable soil.*	**Red**
		2.8 *Building is built on river terraces that can slide or creep.*	**Red**
		2.9 *Building is built on hill slope that can slide.*	**Red**
3.	**Architectural Features**	3.16 *Building has exterior plan dimension smaller at the plinth level than at upper storeys.*	**Red**
		3.17 *Building has heavier upper storeys.*	**Red**
		3.18 *Building has large unanchored projections or overhangs, e.g., parapets.*	**Red**

		3.19 Building has irregular plan geometry and/or plan aspect (length-to-width ratio) >4.0.	**Red**
		3.20 Building has storey height more than about 25 times wall thickness.	**Red**
		3.21 Building has unsupported height more than 2.5m.	**Red**
		3.22 Building has stiff upper storeys.	**Red**
4.	**Structural Aspects**	4.3 Building has no continuous lintel band, roof band and gable band.	**Red**
		4.4 Building has door/window openings in walls close to corners.	**Red**
		4.5 Building has masonry walls not integrated into each other at corners.	**Red**
		4.6 Building has walls made with no or mud mortar exposed to vagaries.	**Red**
		4.7 Building has roof not anchored into the walls.	**Red**
		4.8 Building has thick walls made of two wythes and no through stones.	**Red**
		4.9 Building has out-of-plumb walls.	**Red**
		4.10 Building has openings close to wall junctions or corners, or to floors/roof.	**Red**
		4.11 Building has separation between walls and between walls and floor/roof.	**Red**
		4.12 Building has cracks in walls.	**Red**
5.	**Construction and Maintenance Details**	5.1 Building has walls made with mud mortar and deteriorated and cracked significantly owing to vagaries of outside weather (especially rainwater beating).	**Red**
6.	**Siting Issues**	6.1 Building is resting on cracked ground.	**Yellow**
		6.2 Building is adjoining another building on the side with no gap.	**Yellow**
		6.3 Building is resting on sloped ground with access to house at two/three levels, i.e., ground, middle floor and roof.	**Yellow**
		6.4 Building is connected to the sloped ground and there is no gap between the building and the natural slope of the ground.	**Yellow**
7.	**Soil & Foundation Conditions**	7.1 Building is resting on weak or non-uniform soil along the length in plan.	**Yellow**
8.	**Architectural Features**	8.1 Building has large or heavy cantilever parapet walls.	**Yellow**
		8.2 Building has large or heavy cantilever chimneys.	**Yellow**
		8.3 Building has large or heavy cantilever balconies or sunshades.	**Yellow**
		8.4 Building has large room sizes and large storey heights.	**Yellow**
		8.5 Building has windows larger than 50% of the length of the wall.	**Yellow**
		8.6 Building has unanchored water tanks on roof.	**Yellow**
9.	**Structural Aspects**	9.1 Building has an irregular grid of masonry walls in plan.	**Yellow**
		9.2 Building has staircase at unsymmetrical location in plan.	**Yellow**
		9.3 Building has loosely placed country tiles on the roof.	**Yellow**
10.	**Construction and Maintenance Details**	10.1 Building has vertical cracks in masonry walls.	**Yellow**
		10.2 Building has poor construction quality material.	**Yellow**
11.		None of the above.	Green

FINAL RATING		
GREEN (Usable)	**YELLOW (Usable with temporary interventions)**	**RED (Unusable)**
	Suggested interventions:	

Further Actions		
Building to be quarantined	*: ☐ Not Recommended*	*☐ Recommended*
Level 2 Detailed Visual Screening	*: ☐ Not Recommended*	*☐ Recommended*

.

B: RRSM

Post-Earthquake **Rapid Visual Screening**

Random Rubble Stone Masonry

Inspection

Inspector ID:

Inspection Date :

Inspection Time:

Building Description

Building Name :

Address 1 :

Address 2 :

City/Town :

Coordinates : N_____° E_____°

Structural System

Structural System

Load Bearing Masonry Structure with
- ☐ *Igneous rocks*
- ☐ *Sedimentary rocks*
- ☐ *Slate blocks*
- ☐ *Others _____*

Structural Components

FLOOR	☐ *RC Slab*	☐ *Timber planks on Timber beams*		☐ *Other_____*
ROOF GROMETRY	☐ *Pitched*	☐ *Hipped*	☐ *Split*	☐ *Other_____*
ROOF MATERIAL	☐ *RC Slab*	☐ *Wood Truss with*	○ *Clay tiles*	
			○ *Corrugated sheets*	
			○ *Wood Planks*	
	☐ *Steel Truss with corrugated sheeting*			☐ *Other_____*
MORTAR	☐ *Cement*	☐ *Mud*	☐ *Lime* ☐ *No mortar*	☐ *Other_____*

Occupancy

RESIDENTIAL	☐ *Individual House*	☐ *Apartments*	
EDUCATIONAL	☐ *School*	☐ *College*	☐ *Institute/University*
LIFELINE	☐ *Hospital*	☐ *Police Station*	☐ *Fire Station*
	☐ *Power Station*	☐ *Water Plant*	☐ *Sewage Plant*
COMMERCIAL	☐ *Hotel*	☐ *Shopping*	☐ *Recreational*
OFFICE	☐ *Government*	☐ *Private*	
MIXED USE	☐ *Residential and Commercial*		☐ *Residential and Industrial*
INDUSTRIAL	☐ *Agriculture*	☐ *Live Stock*	
OTHER	☐ _____		

S.No.	Life Threatening Parameters		Tag
1.	**Siting Issues**	1.4 Building is resting on ground that has failed due to Landslide/Fissures and Liquefaction.	**Red**
		1.5 Building is tilted.	**Red**
		1.6 Building is resting on hill slopes or adjacent to hill slopes, and has unsafe/tilted adjoining/uphill building or loose boulders.	**Red**
2.	**Soil & Foundation Conditions**	2.5 Building is resting on river terraces that have cracked or soil.	**Red**
		2.6 Building has sustained uneven settlement of the ground.	**Red**
		2.7 Building is resting on liquefied soil.	**Red**
3.	**Structural Aspects**	5.3 Building has pounding damage from adjoining building or structure.	**Red**
		5.4 Building has collapsed/damaged staircase/mumty or blockade of staircase.	**Red**
		5.5 Building has plinth masonry severely damaged.	**Red**

		5.6 Building has horizontally slid at any storey.	**Red**
		5.7 Building has suffered separation of wall junctions at corners.	**Red**
		5.8 Building has floor-wall junction separated with walls out-of-plumb.	**Red**
		5.9 Building has suffered gable collapse.	**Red**
		5.10 Building has walls separated into wythes.	**Red**
		5.11 Building has diagonal shear cracking in wall piers and/or spandrels.	**Red**
		5.12 Building has crushed masonry at wall base.	**Red**
4.	**Siting Issues**	4.1 Building is adjacent to a failed slope.	**Yellow**
5.	**Structural Aspects**	5.1 Building has sustained collapse of cantilevers, balconies, chimneys and parapets.	**Yellow**
		5.2 Uneven settlement of adjacent building.	**Yellow**
		None of the above	Green

FINAL RATING		
GREEN (Usable)	**YELLOW (Usable with temporary interventions)**	**RED (Unusable)**
	Suggested interventions:	

Further Actions

Building to be quarantined : ☐ Not Recommended ☐ Recommended

Level 2 Detailed Visual Screening : ☐ Not Recommended ☐ Recommended

A: MH

Pre-Earthquake **Rapid Visual Screening**

Mud House

Inspection

Inspector ID:
Inspection Date :
Inspection Time:

Building Description

Building Name :
Address 1 :
Address 2 :
City/Town :
Coordinates : N_____° E_____°

Structural System

Structural System

Load bearing mud walls Structure with
- ☐ *Un-strengthened mud courses*
- ☐ *Slurry of wet mud*
- ☐ *Locally available grass*

Other ☐ _____

Structural Components

FLOOR	☐ *Mud plastered*	☐ *Other*_____	
ROOF GEOMETRY	☐ *Pitched*	☐ *Hipped*	☐ *Other*_____
ROOF MATERIAL	☐ *Thach + Bamboo*	☐ *Wood truss with*	○ *Clay tiles*
			○ *Corrugated sheets*
			○ *Wood planks*

Occupancy

RESIDENTIAL	☐ *Individual House*	☐ *Apartments*	
EDUCATIONAL	☐ *School*	☐ *College*	☐ *Institute/University*
LIFELINE	☐ *Hospital*	☐ *Police Station*	☐ *Fire Station*
	☐ *Power Station*	☐ *Water Plant*	☐ *Sewage Plant*
COMMERCIAL	☐ *Hotel*	☐ *Shopping*	☐ *Recreational*
OFFICE	☐ *Government*	☐ *Private*	
MIXED USE	☐ *Residential and Commercial*		☐ *Residential and Industrial*
INDUSTRIAL	☐ *Agriculture*	☐ *Live Stock*	
OTHER	☐ _____		

S.No.	Life Threatening Parameters		Tag
1.	**Siting Issues**	1.1 *Building is built on hill slope or adjacent to hill slope, that is vulnerable to falling debris.*	**Red**
		1.2 *Building rests adjoining a severely deteriorated or damaged building or structure.*	**Red**
2.	**Soil & Foundation Conditions**	2.1 *Building is built on liquefiable soil.*	**Red**
		2.2 *Building is built on river terraces that can slide or creep.*	**Red**
		2.3 *Building is built on hill slope that can slide.*	**Red**
		2.4 *Building is built on ground with high water table.*	**Red**
3.	**Architectural Features**	3.1 *Building has exterior plan dimension smaller at the plinth level than at upper storeys.*	**Red**
		3.2 *Building has heavier upper storey.*	**Red**
		3.3 *Building has large unanchored projections or overhangs.*	**Red**
		3.4 *Building has irregular plan geometry and/or plan aspect (length-to-width ratio) more than 4.0.*	**Red**

		3.5	*Building has storey height more than about 25 times wall thickness.*	**Red**
		3.6	*Building has unsupported height more than 2.5m.*	**Red**
4.	**Construction Style**	4.1	*Building has no or no continuous lintel band, roof band and gable band.*	**Red**
		4.2	*Building has door/window openings in walls close to corners.*	**Red**
		4.3	*Building has masonry walls not integrated into close to corner.*	**Red**
		4.4	*Building has roof not anchored into the walls.*	**Red**
		4.5	*Building has thick walls made of two wythes and no through mesh of timber.*	**Red**
		4.6	*Building has out-of-plumb walls.*	**Red**
		4.7	*Building has cracks in walls.*	**Red**
		4.8	*Building has no vertical wood posts along the length of the wall.*	**Red**
		4.9	*Building has no wood posts under each rafters of floor/roof.*	**Red**
5.	**Construction and Material Details**	5.1	*Building has mud walls deteriorated and cracked significantly owing to vagaries of the outside weather (especially rainwater beating).*	**Red**
6.	**Siting Issues**	6.1	*Building is resting on cracked ground.*	**Yellow**
		6.2	*Building is adjoining another building on the side with no gap.*	**Yellow**
7.	**Soil & Foundation Conditions**	7.1	*Building is resting on weak or non-uniform soil along the length in plan.*	**Yellow**
8.	**Architectural Features**	8.1	*Building has large room sizes and large storey heights.*	**Yellow**
		8.2	*Building has poor distribution of walls in plan of the building.*	**Yellow**
		8.3	*Building has wall aspect ratio more than 2.*	**Yellow**
		8.4	*Building has windows larger than 50% of the width of the wall.*	**Yellow**
		8.5	*Building has floors with brick, brick coba or mud overlay on (a) Timber joists and planks, and/or (b) Timber planks and stone slabs.*	**Yellow**
9.	**Structural Aspects**	9.1	*Building has an irregular grid of masonry walls in plan.*	**Yellow**
		9.2	*Building has staircase at unsymmetrical location in plan.*	**Yellow**
		9.3	*Building has irregular plan in geometry.*	**Yellow**
10.	**Construction and Maintenance Details**	10.1	*Building has vertical cracks in mud walls.*	**Yellow**
		10.2	*Poor quality of materials.*	**Yellow**
11.		*None of the above.*		**Green**

FINAL RATING		
GREEN (Usable)	**YELLOW (Usable with temporary interventions)**	**RED (Unusable)**
	Suggested interventions:	

Further Actions

Building to be quarantined : ☐ *Not Recommended* ☐ *Recommended*

Level 2 Detailed Visual Screening : ☐ *Not Recommended* ☐ *Recommended*

B: MH

Post-Earthquake **Rapid Visual Screening**

Mud Houses

Inspection

Inspector ID:
Inspection Date :
Inspection Time:

Building Description

Building Name :
Address 1 :
Address 2 :
City/Town :
Coordinates : N_____° E_____°

Structural System

Structural System

Load bearing mud walls Structure with
- ☐ *Un-strengthened mud courses*
- ☐ *Slurry of wet mud*
- ☐ *Locally available grass*

Other ☐ _____

Structural Components

FLOOR	☐ *Mud plastered*	☐ *Other*_____	
ROOF GEOMETRY	☐ *Pitched*	☐ *Hipped*	☐ *Other*_____
ROOF MATERIAL	☐ *Thach + Bamboo*	☐ *Wood truss with*	○ *Clay tiles*
			○ *Corrugated sheets*
			○ *Wood planks*

Occupancy

RESIDENTIAL	☐ *Individual House*	☐ *Apartments*	
EDUCATIONAL	☐ *School*	☐ *College*	☐ *Institute/University*
LIFELINE	☐ *Hospital*	☐ *Police Station*	☐ *Fire Station*
	☐ *Power Station*	☐ *Water Plant*	☐ *Sewage Plant*
COMMERCIAL	☐ *Hotel*	☐ *Shopping*	☐ *Recreational*
OFFICE	☐ *Government*	☐ *Private*	
MIXED USE	☐ *Residential and Commercial*		☐ *Residential and Industrial*
INDUSTRIAL	☐ *Agriculture*	☐ *Live Stock*	
OTHER	☐ _____		

S.No.	Life Threatening Parameters		Tag
1.	**Siting Issues**	1.1 Building is resting on ground that has failed due to Landslide/Fissures and Liquefaction.	**Red**
		1.2 Building is tilted.	**Red**
		1.3 Building is resting on hill slopes or adjacent to hill slopes, and has unsafe/tilted adjoining/uphill building or loose boulders.	**Red**
2.	**Soil & Foundation Conditions**	2.1 Building is resting on river terraces that have cracked or soil.	**Red**
		2.2 Building has sustained uneven settlement of the ground.	**Red**
		2.3 Building is resting on liquefied soil.	**Red**
3.	**Structural Aspects**	3.1 Building has suffered gable collapse.	**Red**
		3.2 Separation of wythes observed.	**Red**
		3.3 Building has suffered damage to masonry plinth.	**Red**
4.	**Siting Issues**	4.1 Building is adjacent to a failed slope.	**Yellow**
5.	**Structural Aspects**	5.1 Building has sustained collapse of cantilevers, balconies, chimneys and parapet.	**Yellow**

		5.2 Uneven settlement of adjacent building.	***Yellow***
6.		*None of the above.*	***Green***

FINAL RATING		
GREEN (Usable)	***YELLOW (Usable with temporary interventions)***	***RED (Unusable)***
	Suggested interventions:	

Further Actions

Building to be quarantined : ☐ *Not Recommended* ☐*Recommended*

Level 2 Detailed Visual Screening : ☐ *Not Recommended* ☐*Recommended*

A:DD

Pre-Earthquake Rapid Visual Screening
Dhajji Diwari

Inspection

Inspector ID:
Inspection Date :
Inspection Time:

Building Description

Building Name :
Address 1 :
Address 2 :
City/Town :
Coordinates : N_____° E_____°

Structural Components

STRUCTURAL SYSTEM
- ☐ *Timber frame without diagonal braces*
- ☐ *Timber frame with diagonal braces* ☐ *Other*_____

FLOOR
- ☐ *Timber beams with Wooden Planks*
- ☐ *Timber frame with stone planks* ☐ *Other*_____

ROOF GEOMETRY ☐ *Flat* ☐ *Pitched* ☐ *Hipped*
- ☐ *Split* ☐ *Other*_____

ROOF MATERIAL
- ☐ *Timber truss with Timber planks*
- ☐ *Timber truss with corrugated GI sheets* ☐ *Other*_____

OTHER ☐ _____

Occupancy

RESIDENTIAL	☐ *Individual House*	☐ *Apartments*	
EDUCATIONAL	☐ *School*	☐ *College*	☐ *Institute/University*
LIFELINE	☐ *Hospital*	☐ *Police Station*	☐ *Fire Station*
	☐ *Power Station*	☐ *Water Plant*	☐ *Sewage Plant*
COMMERCIAL	☐ *Hotel*	☐ *Shopping*	☐ *Recreational*
OFFICE	☐ *Government*	☐ *Private*	
MIXED USE	☐ *Residential and Commercial*		☐ *Residential and Industrial*
INDUSTRIAL	☐ *Agriculture*	☐ *Live Stock*	
OTHER	☐ _____		

S.No.	Life Threatening Parameters		Tag
1.	**Siting**	1.1 *Building is built on hill slope or adjacent to hill slope, that is vulnerable to falling debris.*	**Red**
		1.2 *Building rests adjoining a severely deteriorated or damaged building or structure.*	**Red**
2.	**Soil and foundation conditions**	2.1 *Building is built on liquefiable soil.*	**Red**
		2.2 *Building is built on river terraces that can slide or creep.*	**Red**
		2.3 *Building is built on hill slope that can slide.*	**Red**
3.	**Architecture**	3.1 *Building has exterior plan dimension smaller at the plinth level than at upper storeys.*	**Red**
		3.2 *Building has heavier upper storeys.*	**Red**
		3.3 *Building has large unanchored projections or overhangs.*	**Red**
		3.4 *Building has an Open Storey (at ground /other level).*	**Red**
		3.5 *Building has timber frame with Floating Columns.*	**Red**
		3.6 *Building has discontinuous vertical and horizontal runners in walls.*	**Red**

		3.7 *Building has irregularly spaced grid of vertical and horizontal runners in walls.*	**Red**
		3.8 *Building has irregular plan.*	**Red**
		3.9 *Building has plan aspect (length-to-width) ratio exceeds 4.0.*	**Red**
4.	**Structural**	4.1 *Building has a roof that is not integral.*	**Red**
		4.2 *Building does not rest on stiff ground.*	**Red**
		4.3 *Building has too many panel with no infills between vertical and horizontal runners.*	**Red**
		4.4 *Building has vertical timber members spaced more than 1.5 times the height.*	**Red**
		4.5 *Building has vertical timber members not anchored to plinth masonry.*	**Red**
		4.6 *Building has inadequate mortise and tendon joints.*	**Red**
5.	**Siting**	5.1 *Building is resting on cracked ground.*	**Yellow**
		5.2 *Building is adjoining another building on the side with no gap.*	**Yellow**
6.	**Soil and foundation conditions**	6.1 *Building is resting on weak or non-uniform soil along the length in plan.*	**Yellow**
7.	**Architectural**	7.1 *Building has large or heavy cantilever balconies or sunshades.*	**Yellow**
		7.2 *Building has large room sizes and large storey heights.*	**Yellow**
		7.3 *Building has windows larger than 50% of the length of the wall.*	**Yellow**
		7.4 *Building has more than three storeys.*	**Yellow**
8.	**Structural**	8.1 *Building has an irregular structural grid.*	**Yellow**
		8.2 *Building has braces in vertical plane placed in unsymmetrical locations in plan.*	**Yellow**
		8.3 *Building has staircase at unsymmetrical location in plan.*	**Yellow**
		8.4 *Building has unanchored water tanks on roof.*	**Yellow**
		8.5 *Building has discontinuous vertical and horizontal runners.*	**Yellow**
		8.6 *Building has Jharokhas not anchored to external wood frame of the building.*	**Yellow**
		8.7 *Building has diagonal braces only in upper storeys.*	**Yellow**
9.	**Construction and Material details**	9.1 *Building has loose connections at junctions of vertical and horizontal wood runners.*	**Yellow**
		9.2 *Building has poor construction quality material.*	**Yellow**
10.		*None of the above.*	**Green**

FINAL RATING		
GREEN (Usable)	***YELLOW (Usable with temporary interventions)***	***RED (Unusable)***
	Suggested interventions:	

Further Actions

Building to be quarantined : ☐ *Not Recommended* ☐ *Recommended*

Level 2 Detailed Visual Screening : ☐ *Not Recommended* ☐ *Recommended*

B:DD

Post-Earthquake Rapid Visual Screening
Dhajji Diwari

Inspection

Inspector ID:
Inspection Date :
Inspection Time:

Building Description

Building Name :
Address 1 :
Address 2 :
City/Town :
Coordinates : N_____° E_____°

Structural Components

STRUCTURAL SYSTEM
☐ *Timber frame without diagonal braces*
☐ *Timber frame with diagonal braces* ☐ *Other*_____

FLOOR
☐ *Timber beams with Wooden Planks*
☐ *Timber frame with stone planks* ☐ *Other*_____

ROOF GEOMETRY
☐ *Flat* ☐ *Pitched* ☐ *Hipped*
☐ *Split* ☐ *Other*_____

ROOF MATERIAL
☐ *Timber truss with Timber planks*
☐ *Timber truss with corrugated GI sheets* ☐ *Other*_____

OTHER ☐ _____

Occupancy

RESIDENTIAL	☐ *Individual House*	☐ *Apartments*	
EDUCATIONAL	☐ *School*	☐ *College*	☐ *Institute/University*
LIFELINE	☐ *Hospital*	☐ *Police Station*	☐ *Fire Station*
	☐ *Power Station*	☐ *Water Plant*	☐ *Sewage Plant*
COMMERCIAL	☐ *Hotel*	☐ *Shopping*	☐ *Recreational*
OFFICE	☐ *Government*	☐ *Private*	
MIXED USE	☐ *Residential and Commercial*		☐ *Residential and Industrial*
INDUSTRIAL	☐ *Agriculture*	☐ *Live Stock*	
OTHER	☐ _____		

S.No.	Life Threatening Parameters		Tag
1.	**Siting**	1.1 Building is resting on ground that has failed due to Landslide/Fissures and Liquefaction.	**Red**
		1.2 Building is tilted.	**Red**
		1.3 Building is resting on hill slopes or adjacent to hill slopes.	**Red**
2.	**Soil and foundation conditions**	2.1 Building is resting on river terraces that have cracked or soil.	**Red**
		2.2 Building has sustained uneven settlement of the ground.	**Red**
		2.3 Building is resting on liquefied soil.	**Red**
3.	**Structural**	3.1 Building has separation at junction wood floor to vertical wood frame.	**Red**
		3.2 Building has vertical walls out of plumb.	**Red**
		3.3 Building has suffered gable collapse.	**Red**
		3.4 Building has sustained jharoka collapse.	**Red**
		3.5 Building has suffered extensive damage to masonry plinth.	**Red**
		3.6 Building has suffered horizontal sliding at any storey.	**Red**
4.	**Siting**	4.1 Building is adjacent to a failed slope.	**Yellow**

5.	**Structural**	5.1 *Building has sustained collapse of cantilevers, balconies, chimneys and parapets.*	**Yellow**
		5.2 *Building has water tanks on roof displaced from their supports.*	**Yellow**
		5.3 *Building has sustained extensive separation between wood frame members and infill wall, and has no damage in columns.*	**Yellow**
6.		*None of the above*	**Green**

FINAL RATING		
GREEN (Usable)	**YELLOW (Usable with temporary interventions)**	**RED (Unusable)**
	Suggested interventions:	

Further Actions

Building to be quarantined : ☐ *Not Recommended* ☐ *Recommended*

Level 2 Detailed Visual Screening : ☐ *Not Recommended* ☐ *Recommended*

A: EH

Pre-Earthquake Rapid Visual Screening
Ekra House

Inspection

Inspector ID:
Inspection Date :
Inspection Time:

Building Description

Building Name :
Address 1 :
Address 2 :
City/Town :
Coordinates : N_____° E_____°

Structural System

Structural System

☐ *Timber frame without diagonal braces* ☐ *Timber frame with diagonal braces*

☐ *Other*_____

Structural Components

FLOOR	☐ *Timber beams with wooden planks*	☐ *Timber beams with wooden planks*
	☐ *RC Slab*	☐ *Other*_____
ROOF GEOMETRY	☐ *Pitched*	
ROOF	☐ *Timber Truss with timber planks*	☐ *Timber Truss with Thatch*
	☐ *Timber Truss with Corrugated GI sheets*	☐ *Other*_____
WALLS	☐ *Wood frame infilled with Thatch*	☐ *Wood frame infilled with masonry*
	☐ *Wood frame infilled with bamboo matting*	
	☐ *Wood frame nailed with metal sheeting on vertical wall surface*	
	☐ *Wood frame infilled with Ekra matting and plastered with mud*	
OTHER	☐ _____	

Occupancy

RESIDENTIAL	☐ *Individual House*	☐ *Apartments*	
EDUCATIONAL	☐ *School*	☐ *College*	☐ *Institute/University*
LIFELINE	☐ *Hospital*	☐ *Police Station*	☐ *Fire Station*
	☐ *Power Station*	☐ *Water Plant*	☐ *Sewage Plant*
COMMERCIAL	☐ *Hotel*	☐ *Shopping*	☐ *Recreational*
OFFICE	☐ *Government*	☐ *Private*	
MIXED USE	☐ *Residential and Commercial*		☐ *Residential and Industrial*
INDUSTRIAL	☐ *Agriculture*	☐ *Live Stock*	
OTHER	☐ _____		

S.No.	Life Threatening Parameters		Tag
1.	**Siting Issues**	1.1 *Building is built on hill slope or adjacent to hill slope, that is vulnerable to falling debris.*	**Red**
		1.2 *Building rests adjoining a severely deteriorated or damaged building or structure.*	**Red**
2.	**Soil & Foundation Conditions**	2.1 *Building is built on liquefiable soil.*	**Red**
		2.2 *Building is built on river terraces that can slide or creep.*	**Red**
		2.3 *Building is built on hill slope that can slide.*	**Red**
3.	**Architectural Features**	3.1 *Building built on stilts.*	**Red**
		3.2 *Building has large unanchored projections or overhangs.*	**Red**
		3.3 *Building has no continuous sill or plinth bands.*	**Red**

		3.4 *Building is built adjoining a severely deteriorated or damaged building or structure.*	**Red**
4.	**Structural Aspects**	4.1 *Building has door/window openings close to wall corners.*	**Red**
		4.2 *Building has vertical wood members spaced at more than 1.5 times building height.*	**Red**
		4.3 *Building has vertical wood members not anchored to foundation.*	**Red**
		4.4 *Building has roof that is not anchored into the walls.*	**Red**
		4.5 *Building has walls that are not integrated into each other at the corners.*	**Red**
		4.6 *Building has vertical posts of unequal lengths.*	**Red**
		4.7 *Building has roof rafters and purlins tied to vertical posts with deteriorated choir ropes.*	**Red**
5.	**Siting Issues**	5.1 *Building is resting on cracked ground.*	**Yellow**
		5.2 *Building is adjoining another building on the side with no gap.*	**Yellow**
6.	**Soil & Foundation Conditions**	6.1 *Building is resting on weak or non-uniform soil along the length in plan.*	**Yellow**
7.	**Architectural Features**	7.1 *Building has large or heavy cantilever balconies or sunshades.*	**Yellow**
		7.2 *Building has large room sizes and large storey heights.*	**Yellow**
		7.3 *Building has windows larger than 50% of the length of the wall.*	**Yellow**
		7.4 *Building has more than two storeys.*	**Yellow**
		7.5 *Building has non-uniform location of vertical posts in plan.*	**Yellow**
		7.6 *Houses have small gaps between them.*	**Yellow**
8.	**Structural Aspects**	8.1 *Building has an irregular structural grid.*	**Yellow**
		8.2 *Building has braces in vertical plane placed in unsymmetrical locations in plan.*	**Yellow**
		8.3 *Building has staircase at unsymmetrical location in plan.*	**Yellow**
		8.4 *Building has diagonal braces only in upper storeys.*	**Yellow**
		8.5 *Building has connections using natural materials, like coir ropes and reeds, of poor quality.*	**Yellow**
		8.6 *Building has discontinuous vertical and horizontal runners.*	**Yellow**
9.	**Construction and Maintenance Details**	9.1 *Building has loose connections at junctions of vertical and horizontal wood runners.*	**Yellow**
		9.2 *Building has poor construction quality material.*	**Yellow**
10.		*None of the above.*	*Green*

FINAL RATING		
GREEN (Usable)	**YELLOW (Usable with temporary interventions)**	**RED (Unusable)**
	Suggested interventions:	

Further Actions

Building to be quarantined : ☐ *Not Recommended* ☐ *Recommended*

Level 2 Detailed Visual Screening : ☐ *Not Recommended* ☐ *Recommended*

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B: EH

Post-Earthquake Rapid Visual Screening
Ekra House

Inspection

Inspector ID:
Inspection Date :
Inspection Time:

Building Description

Building Name :
Address 1 :
Address 2 :
City/Town :
Coordinates : N_____° E_____°

Structural System

Structural System

☐ *Timber frame without diagonal braces* ☐ *Timber frame with diagonal braces*

☐ *Other_____*

Structural Components

FLOOR	☐ *Timber beams with wooden planks*	☐ *Timber beams with wooden planks*
	☐ *RC Slab*	☐ *Other_____*
ROOF GEOMETRY	☐ *Pitched*	
ROOF	☐ *Timber Truss with timber planks*	☐ *Timber Truss with Thatch*
	☐ *Timber Truss with Corrugated GI sheets*	☐ *Other_____*
WALLS	☐ *Wood frame infilled with Thatch*	☐ *Wood frame infilled with masonry*
	☐ *Wood frame infilled with bamboo matting*	
	☐ *Wood frame nailed with metal sheeting on vertical wall surface*	
	☐ *Wood frame infilled with Ekra matting and plastered with mud*	
OTHER	☐ _____	

Occupancy

RESIDENTIAL	☐ *Individual House*	☐ *Apartments*	
EDUCATIONAL	☐ *School*	☐ *College*	☐ *Institute/University*
LIFELINE	☐ *Hospital*	☐ *Police Station*	☐ *Fire Station*
	☐ *Power Station*	☐ *Water Plant*	☐ *Sewage Plant*
COMMERCIAL	☐ *Hotel*	☐ *Shopping*	☐ *Recreational*
OFFICE	☐ *Government*	☐ *Private*	
MIXED USE	☐ *Residential and Commercial*		☐ *Residential and Industrial*
INDUSTRIAL	☐ *Agriculture*	☐ *Live Stock*	
OTHER	☐ _____		

S.No.	Life Threatening Parameters		Tag
1.	**Siting Issues**	1.4 Building is resting on ground that has failed due to Landslide/Fissures and Liquefaction.	**Red**
		1.5 Building is tilted.	**Red**
		1.6 Building is resting on hill slopes or adjacent to hill slopes, and has unsafe/tilted adjoining/uphill building or loose boulders.	**Red**
2.	**Soil & Foundation Conditions**	2.1 Building is resting on river terraces that have cracked or soil.	**Red**
		2.2 Building has sustained uneven settlement of the ground.	**Red**
		2.3 Building is resting on liquefied soil.	**Red**
3.	**Structural Aspects**	3.1 Building has separation at junction wood floor to vertical wood frame.	**Red**

		3.2 Building has vertical walls out of plumb.	***Red***
		3.3 Building has suffered gable collapse.	***Red***
		3.4 Building has suffered extensive damage to masonry plinth.	***Red***
		3.5 Building has suffered horizontal sliding at any storey.	***Red***
		3.6 Building has differential settlement of vertical posts.	***Red***
4.	***Siting Issues***	*4.1 Building is adjacent to a failed slope.*	***Yellow***
5.	***Structural Aspects***	*5.1 Building has sustained collapse of cantilevers, balconies, chimneys and parapets.*	***Yellow***
		5.2 Building has sustained extensive separation between wood frame members and infill wall, and has no damage in columns.	***Yellow***
6.		*None of the above*	***Green***

FINAL RATING

GREEN (Usable)	***YELLOW (Usable with temporary interventions)***	***RED (Unsable)***
	Suggested interventions:	

Further Actions

Building to be quarantined : ☐ *Not Recommended* ☐ *Recommended*

Level 2 Detailed Visual Screening : ☐ *Not Recommended* ☐ *Recommended*